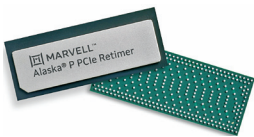




is crucial for reliable communication within AI servers, between board-mounted GPUs, or between CPUs and CXL-enabled shared memory pools.



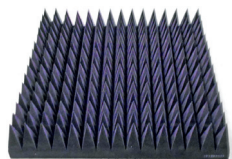
They can be integrated into AI accelerator baseboards, server motherboards, riser cards, or used in active electrical and optical cables for multi-rack server architectures. The retimers feature the industry's lowest power consumption at 10 watts for the 5nm 16-lane PCIe Gen 6 version and offer advanced features such as low-latency modes for cache-coherent links, extensive telemetry, and diagnostics tools.

Marvell Technology

<https://www.marvell.com>

Environmentally sustainable radio wave absorbers

TDK Corporation has launched its new radio wave absorber, the IS-BP series, which incorporates over 25 wt% of biomass material. This sustainable alternative, made primarily from



sugarcane-derived polyethylene, supports CO₂ reduction efforts.

The series is designed for microwave and EMC anechoic chambers that test antennas, wireless communication equipment, and millimetre-wave radar for AD and ADAS. The series has been certified by the JBPA and offers a 13% reduction in CO₂ emissions compared to traditional materials. This series is intended for use in a variety of settings, including the automotive, ICT, and industrial sectors, where it contributes to sustainable practices.

TDK Corporation

<https://www.tdk.com>

Chips for longer battery life

Intel has launched 'Lunar Lake' processors, aimed at enhancing machine

learning and artificial intelligence capabilities on laptops and compact computing systems. The chips feature a 48 TOPs neural coprocessor and offer



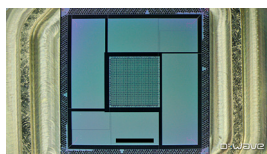
a 40% reduction in power usage compared to previous generations. The Lunar Lake architecture promises a 40% improvement in energy efficiency and a 60% increase in real-world battery performance. This is supplemented by up to 67 TOPs from the GPU. The processor architecture also includes a 'platform controller tile,' which enhances system security and connectivity. Additionally, up to 32GB of on-package memory is included to reduce latency and lower overall power consumption.

Intel

<https://www.intel.com>

Fast anneal feature enhances quantum computing

D-Wave Quantum Inc. has launched the fast-anneal feature on all its quantum processing units (QPUs) available through the Leap real-time quantum



cloud service. This feature speeds up quantum computations by minimising disruptions from thermal fluctuations and noise, which are common barriers in quantum calculations. It enables users to achieve quicker annealing times. The feature is expected to attract interest from both commercial and academic sectors, aiming to advance applications, benchmarking studies, and connect increased coherence to improved performance. The company claims that the fast-anneal feature can enhance the development of quantum-powered generative AI models, accelerating drug discovery and material design through efficient encoding of complex data patterns.

D-Wave Quantum

<https://www.dwavesys.com>

High-performance compute chiplet

Alphawave Semi has teamed up with ARM to develop a high-performance compute chiplet using Arm Neoverse Compute Subsystems for AI/ML, high-performance computing, data centres,



and 5G/6G networks. Built on leading process nodes, this

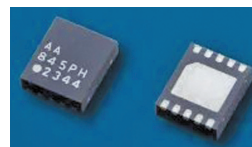
technology helps customers develop and launch advanced SoCs quickly, catering to high-performance digital infrastructure. The company claims to bring advanced packaging and high-speed connectivity technologies, such as PCIe Gen 6.0 and 7.0, Universal Chiplet Express, Ethernet, and HBM subsystems, ensuring powerful and flexible performance for the next-generation infrastructure while improving time-to-market.

Alphawave Semi

<https://awavesemi.com>

A code-free 1-coil fan driver

Melexis has introduced the MLX90418, a 1-coil BLDC fan driver that simplifies server cooling systems. This integrated circuit is the first to offer a code-free, single-coil fan driver



capable of performing specialised server functions such as power-

loss braking and managing AC power losses. It is designed for the server market, reducing bill of material (BoM) costs by up to 25% compared to three-phase BLDC fan systems. The MLX90418 extends the range to 30W for 4cm and 6cm server fans. Its plug-and-play feature eliminates the need for software validation required by MCU-based solutions, saving engineering time and enabling quicker production scaling.

Melexis

<https://www.melexis.com>